

NAT Lab

The screenshot shows a Wireshark capture of network traffic. The filter is set to 'http'. The packet list shows a POST request from 192.168.1.100 to 74.125.91.113. The packet details pane shows the Ethernet II header, Internet Protocol Version 4 header, and the Hypertext Transfer Protocol (HTTP) section. The HTTP section shows a POST request to /safebrowsing/downloads?client=navclient-auto-ffox&appver=... with a status of 200 OK.

No.	Time	Source	Destination	Protocol	Length	Info
7	1.208840	192.168.1.100	74.125.91.113	HTTP	1035	POST /safebrowsing/downloads?client=navclient-auto-ffox&appver=...
11	1.274062	74.125.91.113	192.168.1.100	HTTP	853	HTTP/1.1 200 OK (application/vnd.google.safebrowsing-update)
20	1.572315	192.168.1.100	74.125.106.31	HTTP	767	GET /safebrowsing/rd/goog-malware-shavar_s_15361-15365.15361-15...
39	1.677779	74.125.106.31	192.168.1.100	HTTP	651	HTTP/1.1 200 OK (application/vnd.google.safebrowsing-chunk)
41	1.976996	192.168.1.100	74.125.106.31	HTTP	772	GET /safebrowsing/rd/goog-malware-shavar_a_14466-14470.14466.14...
42	2.000629	74.125.106.31	192.168.1.100	HTTP	881	HTTP/1.1 200 OK (application/vnd.google.safebrowsing-chunk)
43	2.014105	192.168.1.100	74.125.106.31	HTTP	776	GET /safebrowsing/rd/goog-phish-shavar_s_48291-48300.48291-4829...
44	2.038247	74.125.106.31	192.168.1.100	HTTP	526	HTTP/1.1 200 OK (application/vnd.google.safebrowsing-chunk)
45	2.044751	192.168.1.100	74.125.106.31	HTTP	776	GET /safebrowsing/rd/goog-phish-shavar_a_67721-67760.67721-6772...
46	2.064877	74.125.106.31	192.168.1.100	HTTP	1089	HTTP/1.1 200 OK (application/vnd.google.safebrowsing-chunk)
56	7.109267	192.168.1.100	64.233.169.104	HTTP	689	GET / HTTP/1.1
60	7.158797	64.233.169.104	192.168.1.100	HTTP	814	HTTP/1.1 200 OK (text/html)
62	7.281399	192.168.1.100	64.233.169.104	HTTP	719	GET /intl/en_ALL/images/logo.gif HTTP/1.1
73	7.349451	64.233.169.104	192.168.1.100	HTTP	226	HTTP/1.1 200 OK (GIF89a)
75	7.370185	192.168.1.100	64.233.169.104	HTTP	809	GET /extern_js/f/CgJlbhICdXMrMao4NUAILCswDjgHLCswFjgQLCswFzgL...
92	7.448649	64.233.169.104	192.168.1.100	HTTP	648	HTTP/1.1 200 OK (text/javascript)
94	7.492324	192.168.1.100	64.233.169.104	HTTP	695	GET /extern_chrome/ee36edbd3c16a1c5.js HTTP/1.1
100	7.537353	64.233.169.104	192.168.1.100	HTTP	870	HTTP/1.1 200 OK (text/html)
104	7.573305	192.168.1.100	74.125.91.113	HTTP	709	GET /generate_204 HTTP/1.1
106	7.631819	74.125.91.113	192.168.1.100	HTTP	179	HTTP/1.1 204 No Content

Frame 7: Packet, 1035 bytes on wire (8280 bits), 1035 bytes captured (8280 bits) on interface 0

Ethernet II, Src: HonHaiPrecis_0d:ca:8f (00:22:68:0d:ca:8f), Dst: Ciscosys_45:1f:1b (00:22:6b:45:1f:1b)

Destination: Ciscosys_45:1f:1b (00:22:6b:45:1f:1b)

Source: HonHaiPrecis_0d:ca:8f (00:22:68:0d:ca:8f)

Type: IPv4 (0x0800)

Internet Protocol Version 4, Src: 192.168.1.100, Dst: 74.125.91.113

Hypertext Transfer Protocol: Protocol

Packets: 134 · Displayed: 26 (19.4%)

1. IP: 192.168.1.100
- 2.

The screenshot shows a Wireshark capture of network traffic. The filter is set to 'http && ip.addr == 64.233.169.104'. The packet list shows a GET request from 192.168.1.100 to 64.233.169.104. The packet details pane shows the Ethernet II header, Internet Protocol Version 4 header, and the Hypertext Transfer Protocol (HTTP) section. The HTTP section shows a GET request to /safebrowsing/downloads?client=navclient-auto-ffox&appver=... with a status of 200 OK.

No.	Time	Source	Destination	Protocol	Length	Info
56	7.109267	192.168.1.100	64.233.169.104	HTTP	689	GET / HTTP/1.1
60	7.158797	64.233.169.104	192.168.1.100	HTTP	814	HTTP/1.1 200 OK (text/html)
62	7.281399	192.168.1.100	64.233.169.104	HTTP	719	GET /intl/en_ALL/images/logo.gif HTTP/1.1
73	7.349451	64.233.169.104	192.168.1.100	HTTP	226	HTTP/1.1 200 OK (GIF89a)
75	7.370185	192.168.1.100	64.233.169.104	HTTP	809	GET /extern_js/f/CgJlbhICdXMrMao4NUAILCswDjgHLCswFjgQLCswFzgL...
92	7.448649	64.233.169.104	192.168.1.100	HTTP	648	HTTP/1.1 200 OK (text/javascript)
94	7.492324	192.168.1.100	64.233.169.104	HTTP	695	GET /extern_chrome/ee36edbd3c16a1c5.js HTTP/1.1
100	7.537353	64.233.169.104	192.168.1.100	HTTP	870	HTTP/1.1 200 OK (text/html)
107	7.652836	192.168.1.100	64.233.169.104	HTTP	712	GET /images/nav_logo7.png HTTP/1.1
112	7.682361	192.168.1.100	64.233.169.104	HTTP	806	GET /csi?v=3&s=webhp&action=&tran=undefined&e=17259,21588,21766,21920...
119	7.685786	64.233.169.104	192.168.1.100	HTTP	1359	HTTP/1.1 200 OK (PNG)
122	7.709490	192.168.1.100	64.233.169.104	HTTP	670	GET /favicon.ico HTTP/1.1
124	7.737783	64.233.169.104	192.168.1.100	HTTP	269	HTTP/1.1 204 No Content
127	7.763501	64.233.169.104	192.168.1.100	HTTP	1204	HTTP/1.1 200 OK (image/x-icon)

Frame 56: Packet, 689 bytes on wire (5512 bits), 689 bytes captured (5512 bits) on interface 0

Ethernet II, Src: HonHaiPrecis_0d:ca:8f (00:22:68:0d:ca:8f), Dst: Ciscosys_45:1f:1b (00:22:6b:45:1f:1b)

Destination: Ciscosys_45:1f:1b (00:22:6b:45:1f:1b)

Source: HonHaiPrecis_0d:ca:8f (00:22:68:0d:ca:8f)

Type: IPv4 (0x0800)

Internet Protocol Version 4, Src: 192.168.1.100, Dst: 64.233.169.104

NAT home side.pcap

Packets: 134 · Displayed: 14 (10.4%)

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3. Source IP: 192.168.1.100
Destination IP: 64.233.169.104
Source port: 4335
Destination port: 80
4. Source IP: 64.233.169.104
Destination IP: 192.168.1.100
Source port: 80
Destination port: 4335

Wireshark packet capture of NAT traffic. The packet list shows a SYN packet (53) from 192.168.1.100 to 64.233.169.104, a SYN-ACK packet (54) from 64.233.169.104 to 192.168.1.100, and an ACK packet (55) from 192.168.1.100 to 64.233.169.104. The packet details for packet 53 are expanded, showing TCP flags and sequence numbers.

No.	Time	Source	Destination	Protocol	Length	Info
53	7.075657	192.168.1.100	64.233.169.104	TCP	66	4335 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=4 SACK_PERM
54	7.108986	64.233.169.104	192.168.1.100	TCP	66	80 → 4335 [SYN, ACK] Seq=0 Ack=1 Win=5720 Len=0 MSS=1430 SACK_P...
55	7.109053	192.168.1.100	64.233.169.104	TCP	54	4335 → 80 [ACK] Seq=1 Ack=1 Win=260176 Len=0

Identification: 0xa2ab (41643)
 010. = Flags: 0x2, Don't fragment
 ...0 0000 0000 0000 = Fragment Offset: 0
 Time to Live: 128
 Protocol: TCP (6)
 Header Checksum: 0xabc6 [validation disabled]
 [Header checksum status: Unverified]
 Source Address: 192.168.1.100
 Destination Address: 64.233.169.104
 [Stream index: 5]
 Transmission Control Protocol, Src Port: 4335, Dst Port: 80, Seq: 1, Ack: 1, Source Port: 4335

5. TCP SYN (packet 53) at t=7.075657
 - a. Source IP: 192.168.1.100
 - b. Destination IP: 64.233.169.104
 - c. Source port: 4335
 - d. Destination port: 80
 - TCP SYN-ACK (packet 54) at t=7.108986
 - a. Source IP: 64.233.169.104
 - b. Destination IP: 192.168.1.100
 - c. Source port: 80
 - d. Destination port: 4335
 - TCP ACK (packet 55) at t=7.109053
 - a. Source IP: 192.168.1.100
 - b. Destination IP: 64.233.169.104
 - c. Source port: 4335
 - d. Destination port: 80
- The ACK was received at the client at t=7.109053.

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The image shows a Wireshark capture of network traffic. The top pane displays a list of packets, with packet 85 selected. The middle pane shows the details of the selected packet, which is an HTTP GET request. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
85	6.069168	71.192.34.104	64.233.169.104	HTTP	689	GET / HTTP/1.1
90	6.117570	64.233.169.104	71.192.34.104	HTTP	814	HTTP/1.1 200 OK (text/html)
93	6.241357	71.192.34.104	64.233.169.104	HTTP	719	GET /intl/en_ALL/images/logo.gif HTTP/1.1
103	6.308118	64.233.169.104	71.192.34.104	HTTP	226	HTTP/1.1 200 OK (GIF89a)
106	6.330131	71.192.34.104	64.233.169.104	HTTP	809	GET /extern_js/f/Cg1lbhICdXMrMAo4NUAILCswDjgHLCswFjgQLCswFzgLCSwG0gE...
121	6.407366	64.233.169.104	71.192.34.104	HTTP	648	HTTP/1.1 200 OK (text/javascript)
125	6.452270	71.192.34.104	64.233.169.104	HTTP	695	GET /extern_chrome/ee36edbd3c16a1c5.js HTTP/1.1
131	6.496234	64.233.169.104	71.192.34.104	HTTP	870	HTTP/1.1 200 OK (text/html)
139	6.612801	71.192.34.104	64.233.169.104	HTTP	712	GET /images/nav_logo7.png HTTP/1.1
144	6.642308	71.192.34.104	64.233.169.104	HTTP	806	GET /csi?v=3&s=webhp&action=&tran=undefined&e=17259,21588,21766,21920...
149	6.644609	64.233.169.104	71.192.34.104	HTTP	1359	HTTP/1.1 200 OK (PNG)
154	6.669397	71.192.34.104	64.233.169.104	HTTP	670	GET /favicon.ico HTTP/1.1
157	6.696669	64.233.169.104	71.192.34.104	HTTP	269	HTTP/1.1 204 No Content
160	6.722203	64.233.169.104	71.192.34.104	HTTP	1204	HTTP/1.1 200 OK (image/x-icon)

Details of packet 85 (HTTP GET):

- ... 0101 = Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 675
- Identification: 0xa2ac (41644)
- 010. = Flags: 0x2, Don't fragment
- ... 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 127
- Protocol: TCP (6)
- Header Checksum: 0x022f [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 71.192.34.104
- Destination Address: 64.233.169.104

Raw packet data (hex):

```

0020 a9 68 10 ef 00 50 f8 32 36 e5 e9 4f 38 95 50 18 h.P:2 6..08.P.
0030 fe 14 38 6d 00 00 47 45 54 20 2f 20 48 54 50 ..8m..GE T / HTTP
0040 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 77 77 77 2e /1.1 Ho st: www.
0050 67 6f 6f 6f 6c 65 2e 63 6f 6d 0d 0a 55 73 65 72 google.c om User
0060 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 2f -Agent: Mozilla/
0070 35 2e 30 20 28 57 69 6e 64 6f 77 73 3b 20 55 3b 5.0 (Win dows; U;
0080 20 57 69 6e 64 6f 77 73 20 4e 54 20 35 2e 31 3b Windows NT 5.1;
0090 20 65 6e 2d 55 53 3b 20 72 76 3a 31 2e 39 2e 30 en-US; rv:1.9.0
00a0 2e 31 34 29 20 47 65 63 6b 6f 2f 32 30 30 30 30 .14) gec ko/20090
00b0 38 32 37 30 37 20 46 69 72 65 66 6f 78 2f 33 2e 82707 Fi refox/3.
00c0 30 2e 31 34 20 28 2e 4e 45 54 20 43 4c 52 20 33 0.14 (.N ET CLR 3
00d0 2e 35 2e 33 30 37 32 39 29 0d 0a 41 63 63 65 70 5.30729 )- Accep
00e0 74 3a 20 74 65 78 74 2f 68 74 6d 6c 2c 61 70 70 t: text/ html,app
00f0 6c 69 63 61 74 69 6f 6e 2f 78 68 74 6d 6c 2b 78 lication /xhtml+x
  
```

6. The HTTP GET appears in NAT_ISP_side at t = 6.069168, which is earlier than t = 7.109267 in home_side, as expected.

Comparison of Traces

Field	NAT_home_side (Q3)	NAT_ISP_side
Source IP	192.168.1.100	71.192.34.104 ← changed
Destination IP	64.233.169.104	64.233.169.104 ← same
Source Port	4335	4335
Destination Port	8080	8080
Key Observation		

The IP address 71.192.34.104 is the router's public IP address.

This shows that NAT (Network Address Translation) replaced the private client IP (192.168.1.100) with the public IP before sending the packet to the ISP side.

7.

IP Field	Changed?	Why
Version	No	Still IPv4
Header Length	No	Same structure
Flags	No	Unchanged
Checksum	Yes	Source IP was modified, so the checksum had to be recomputed

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Only the checksum field changes in this case because any modification to the IP header (such as NAT changing the source IP address) requires the checksum to be recalculated. All other fields remain unchanged.

8. The 200 OK appears in NAT_ISP_side at t = 6.117570 (packet 90).

Comparison of Traces

Field	NAT_home_side (Q4)	NAT_ISP_side	Same / Different
Source IP	64.233.169.104	64.233.169.104	Same
Destination IP	192.168.1.100	71.192.34.104	Different
Source Port	8080	8080	Same
Destination Port	4335	4335	Same

The destination IP changes from the client's private IP (192.168.1.100) to the router's public IP (71.192.34.104).

This shows that NAT is handling return traffic by routing it back through the router, which then forwards it to the correct internal device.

The image shows a Wireshark packet capture of NAT_ISP_side.pcap. The filter is set to tcp.port == 4335. The packet list shows a GET request (packet 85) and a 200 OK response (packet 90). The packet details pane for packet 85 shows the HTTP GET request to /int1/en_ALL/images/logo.gif. The packet bytes pane shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
82	6.035475	71.192.34.104	64.233.169.104	TCP	66	66 4335 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=4 SACK_PERM
83	6.067775	64.233.169.104	71.192.34.104	TCP	66	80 → 4335 [SYN, ACK] Seq=0 Ack=1 Win=5720 Len=0 MSS=1430 SACK_P...
84	6.068754	71.192.34.104	64.233.169.104	TCP	60	60 4335 → 80 [ACK] Seq=1 Ack=1 Win=260176 Len=0
85	6.069168	71.192.34.104	64.233.169.104	HTTP	689	GET / HTTP/1.1
87	6.099637	64.233.169.104	71.192.34.104	TCP	60	80 → 4335 [ACK] Seq=1 Ack=636 Win=7040 Len=0
88	6.117078	64.233.169.104	71.192.34.104	TCP	1484	80 → 4335 [ACK] Seq=1 Ack=636 Win=7040 Len=1430 [TCP PDU reasse...
89	6.117407	64.233.169.104	71.192.34.104	TCP	1484	80 → 4335 [ACK] Seq=1431 Ack=636 Win=7040 Len=1430 [TCP PDU rea...
90	6.117570	64.233.169.104	71.192.34.104	HTTP	814	HTTP/1.1 200 OK (text/html)
91	6.118515	71.192.34.104	64.233.169.104	TCP	60	4335 → 80 [ACK] Seq=636 Ack=3621 Win=260176 Len=0
93	6.241357	71.192.34.104	64.233.169.104	HTTP	719	GET /int1/en_ALL/images/logo.gif HTTP/1.1
94	6.273849	64.233.169.104	71.192.34.104	TCP	309	80 → 4335 [PSH, ACK] Seq=3621 Ack=1301 Win=8320 Len=255 [TCP PD...
95	6.274230	64.233.169.104	71.192.34.104	TCP	1484	80 → 4335 [ACK] Seq=3876 Ack=1301 Win=8320 Len=1430 [TCP PDU re...
96	6.274571	64.233.169.104	71.192.34.104	TCP	1484	80 → 4335 [ACK] Seq=5306 Ack=1301 Win=8320 Len=1430 [TCP PDU re...
97	6.274853	64.233.169.104	71.192.34.104	TCP	1290	80 → 4335 [PSH, ACK] Seq=6736 Ack=1301 Win=8320 Len=1236 [TCP P...
98	6.275315	71.192.34.104	64.233.169.104	TCP	60	4335 → 80 [ACK] Seq=1301 Ack=5306 Win=260176 Len=0
99	6.275965	71.192.34.104	64.233.169.104	TCP	60	4335 → 80 [ACK] Seq=1301 Ack=7972 Win=260176 Len=0
100	6.307419	64.233.169.104	71.192.34.104	TCP	1484	80 → 4335 [ACK] Seq=7972 Ack=1301 Win=8320 Len=1430 [TCP PDU re...
101	6.307738	64.233.169.104	71.192.34.104	TCP	1484	80 → 4335 [ACK] Seq=9402 Ack=1301 Win=8320 Len=1430 [TCP PDU re...
102	6.308043	64.233.169.104	71.192.34.104	TCP	1484	80 → 4335 [ACK] Seq=10832 Ack=1301 Win=8320 Len=1430 [TCP PDU r...
103	6.308118	64.233.169.104	71.192.34.104	HTTP	226	HTTP/1.1 200 OK (GIF89a)

9. SYN Packet Comparison

Field	NAT_home_side (Q5)	NAT_ISP_side	Same / Different
SYN Source IP	192.168.1.100	71.192.34.104	Different
SYN Destination IP	64.233.169.104	64.233.169.104	Same
SYN Source Port	4335	4335	Same
SYN Destination Port	80	80	Same

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SYN Time 7.075657 6.035475 Different

ACK Packet Comparison

Field NAT_home_side (Q5) NAT_ISP_side Same / Different

ACK Source IP 192.168.1.100 71.192.34.104 Different

ACK Destination IP 64.233.169.104 64.233.169.104 Same

ACK Time 7.109053 6.068754 Different

10. NAT Translation Table

WAN Side LAN Side

71.192.34.104 : 4335 192.168.1.100 : 4335

In this case, the NAT router only needed to translate the **IP address**, while the **port number remained unchanged (4335)**.